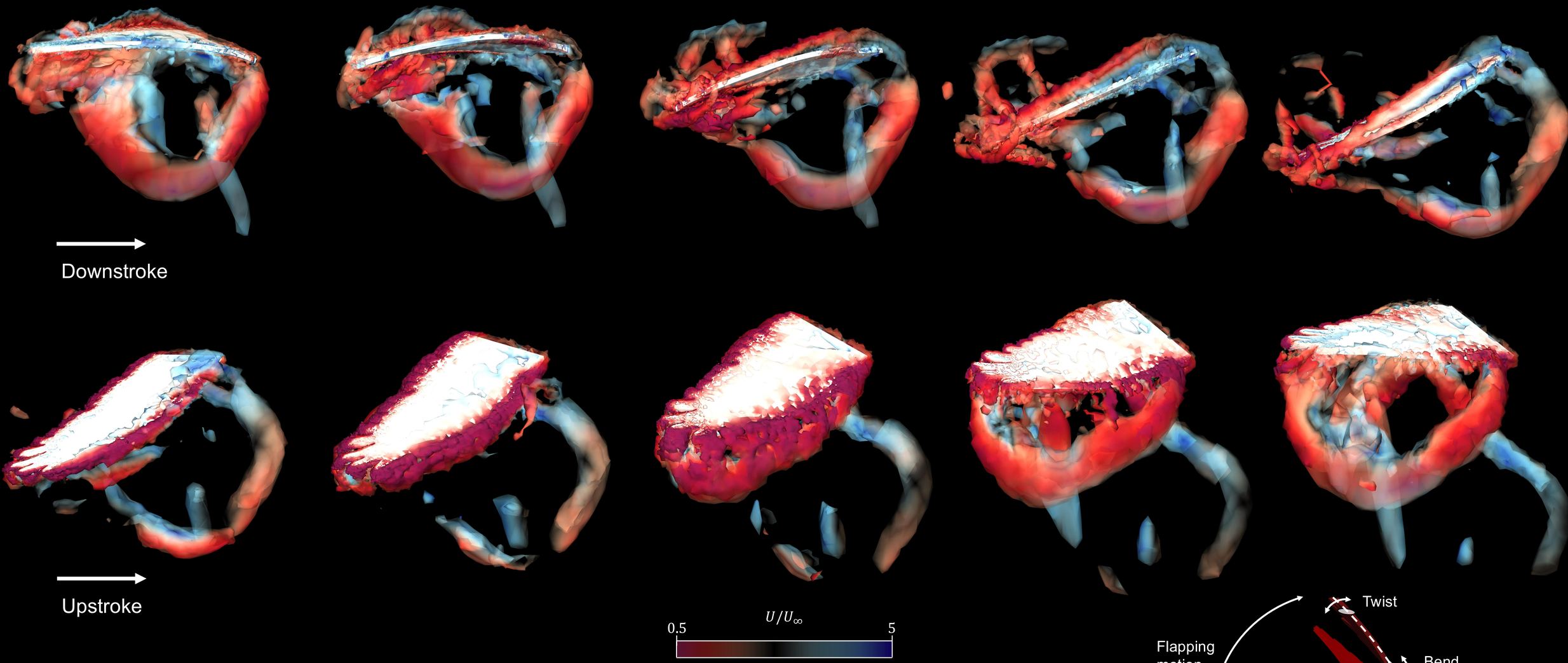
Wake Dynamics of An Actively Deforming Owl Wing

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Bioinspired Fluid-Structure Interaction Laboratory



An owl-inspired wing of constant thickness sweeps through a coupled sinusoidal motion of flapping, pitching, twisting, and bending. Over one flapping cycle, leading-edge separation rolls into a coherent vortex that grows over the wing, connects with the tip vortex, and develops into a striking three-dimensional arch. As the wing reverses direction, the arch evolves into a loop-like structure that progressively detaches and convects into the wake, while a slender vortical connection persists through much of the return stroke. The shed loop stretches and drifts away as the next vortex begins to form. Q-criterion iso-surfaces, coloured by normalised velocity magnitude, reveal the evolving three-dimensional architecture of the wake.

